Hypertext Preprocessor (PHP) (Part1)

MMI230702 - Week 6

PHP - Parts 1

Welcome to this week's practical session, which is designed to get you familiarized with PHP—a powerful server-side scripting language used to create dynamic web pages. PHP stands for "Hypertext Pre-processor" and is widely used for web development due to its flexibility and ease of integration with HTML.

What you'll learn:

1. **Introduction to PHP:** Understand the basics of PHP and its importance in web development.
2. **Simple PHP Syntax:** Learn the basic syntax of PHP and how to embed PHP code within HTML.
3. **Creating a Simple Webpage:** Practice creating a basic webpage using PHP tags.

Creating a Simple Webpage with PHP:

To create a simple webpage with PHP, you need to have a server that can process PHP scripts (like XAMPP, WAMP, or MAMP). Here's a basic example of how you can mix PHP with HTML to display a message:

```
<!DOCTYPE html>

<html>

<head>

  <title>My First PHP Page</title>

</head>

<body>

  <h1>Welcome to My PHP Page</h1>

  <?php

    // This is a PHP comment

    echo "<p>Hello, World! This is my first PHP script.</p>";

  ?>

</body>

</html>
```

In the example above:

- The PHP code is embedded within the <?php ...?> tags.
- The echo statement is used to output text to the webpage.
- PHP comments start with // and are used to annotate the code.

Practical Exercises: Choose any of the exercises provided and start coding. These exercises are designed to help you apply the concepts you've learned and gain hands-on experience with PHP. Don't hesitate to ask for help if you're stuck.

1. Figure out what this program prints then create a PHP page with the code and run it to see if it prints what you expected.

```
$age = 12;
$shoe_size = 13;
if ($age > $shoe_size)
{
        print "Message 1.";
}
elseif (($shoe_size++) && ($age > 20))
{
        print "Message 2.";
}
else
{
        print "Message 3.";
}
print "Age: $age. Shoe Size: $shoe_size";
```

2. Write a PHP page to output the sequence "1 2 4 8 16......1024" all on one line, using a for loop. (You can use **print(i);** to output the value of the variable i). Note that a variable in PHP is prefixed with a $ sign.

3. Write a PHP page to output the sequence "1 2 4 8 16......1024" on separate lines, using a for loop. (Hint, you can output html from within PHP using (for example) **print("
");**)

4. Write a PHP page to output the following table using a for loop.

1
2
4
8
16
32
64
128
256
512
1024

5. Write a PHP page to output the following table using nested for loops.

1	2	3	4	5
2	4	6	8	10
3	6	9	12	15
4	8	12	16	20
5	10	15	20	25

6. Rewrite the 5 by 5 table above to print consecutive numbers.

1	2	3	4	5
6	7	8	9	10
11	12	13	14	15
16	17	18	19	20
21	22	23	24	25

7. Use a **while loop** to print out a table of Fahrenheit and Celsius temperature equivalents from -50 degrees F to 50 degrees F in 5-degree increments. On the Fahrenheit temperature scale, water freezes at 32 degrees and boils at 212 degrees. On the Celsius scale, water freezes at 0 degrees and boils at 100 degrees. So, to convert from Fahrenheit to Celsius, you subtract 32 from the temperature, multiply by 5, and divide by 9. To convert from Celsius to Fahrenheit, you multiply by 9, divide by 5, and then add 32.

8. Modify your answer to the above question to use for () instead of while ().

9. Write a script that creates a variable and assigns an integer value to it, then adds 1 to the variable's value three times, using a different operator each time. Display the final result to the user.

10. Write a script that creates two variables and assigns a different integer value to each variable. Now make your script test whether the first value is

    a. equal to the second value
    b. greater than the second value
    c. less than or equal to the second value
    d. not equal to the second value

Output the result of each test to the user.